MRA Project Part A

USER NAME: RAJESH MAMBULLI

BATCH: **PGPDSBA.O.SEP24.A**

EMAIL: RAJESHMAMBULLI@ICLOUD.COM

Agenda – Part A: Customer Behavior Analysis & Segmentation Strategy

- **Business Overview:** Introduction to the company and the analytical challenge.
- **Problem Statement & Objectives:** Key questions and business goals driving the analysis
- **Data Understanding:** Structure and variables of the transactional dataset
- **Exploratory Data Analysis (EDA):** Key patterns, sales trends, and product-level insights
- **Insights from EDA** Highlights of customer behavior and potential segmentation cues
- **Customer Segmentation (RFM Analysis):** Methodology for RFM scoring and segment creation
- **Segment Profiles & Behavioral Inference:** Interpretation of segments and their business implications
- **Targeted Marketing Strategy:** Actionable recommendations tailored to each segment
- **KNIME Workflow Overview:** Analytical pipeline and tools used for implementation
- **Conclusion & Strategic Impact:** Summary of insights, projected business benefits, and future scope

Business Overview

An automobile parts manufacturing company has been actively selling products to a diverse range of customers for the past three years. Despite its growth, the company lacks the in-house expertise to derive actionable insights from its transaction data. As a result, they wish to uncover hidden patterns and trends in their customer transactions. By analyzing this data, the company aims to better understand customer behavior, improve customer segmentation, and implement targeted marketing strategies. These insights will help the company not only enhance customer satisfaction but also drive revenue growth by offering more personalized and efficient services.

Problem Statement

Objective:

The primary objective of this analysis is to leverage data science techniques to:

Identify underlying patterns in customer purchasing behavior.

Segment customers based on their transactional data.

Provide actionable insights to optimize the company's marketing efforts.

Recommend personalized marketing strategies for each customer segment to maximize sales and customer retention.

Your role as a Business Analyst is to use the provided dataset to achieve these goals and present findings in a manner that can guide the company's decision-making.

Data Description

The dataset provided contains three years of transactional data from the company, with each row representing a unique order. Below is an explanation of the key attributes:

ORDERNUMBER: Unique identifier for each order.
QUANTITYORDERED: Number of items ordered in a specific transaction.
PRICEEACH: Price per unit of the product in the order.
ORDERLINENUMBER: Sequence number of the product in the order.
SALES: Total sales value for the order.
ORDERDATE: Date when the order was placed.
DAYS_SINCE_LASTORDER: Number of days since the customer's previous order.
STATUS: Current status of the order (e.g., Shipped, Disputed).
PRODUCTLINE: Product category to which the item belongs (e.g., Motorcycles, Classic Cars).
MSRP: Manufacturer’s Suggested Retail Price for the product.
PRODUCTCODE: Unique identifier for the product.
CUSTOMERNAME: Name of the customer placing the order.
PHONE: Customer's contact phone number.
ADDRESSLINE1: Customer's primary address.
CITY: City of the customer's address.
POSTALCODE: Postal code of the customer's address.
COUNTRY: Country of the customer's address.
CONTACTLASTNAME: Last name of the customer’s contact person.
CONTACTFIRSTNAME: First name of the customer’s contact person.
DEALSIZE: Size category of the transaction (e.g., Small, Medium, Large).

EDA STARTS

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDERDATE	DAYS_SINCE_LASTORDER	STATUS	PRODUCTLINE	MSRP	PRODUCTC
0	10107	30	95.700	2	2871.000	2018-02-24	828	Shipped	Motorcycles	95	S10_1
1	10121	34	81.350	5	2765.900	2018-05-07	757	Shipped	Motorcycles	95	S10_1
2	10134	41	94.740	2	3884.340	2018-07-01	703	Shipped	Motorcycles	95	S10_1
3	10145	45	83.260	6	3746.700	2018-08-25	649	Shipped	Motorcycles	95	S10_1
4	10168	36	96.660	1	3479.760	2018-10-28	586	Shipped	Motorcycles	95	S10_1
5	10180	29	86.130	9	2497.770	2018-11-11	573	Shipped	Motorcycles	95	S10_1
6	10188	48	114.840	1	5512.320	2018-11-18	567	Shipped	Motorcycles	95	S10_1
7	10211	41	114.840	14	4708.440	2019-01-15	510	Shipped	Motorcycles	95	S10_1
8	10223	37	107.180	1	3965.660	2019-02-20	475	Shipped	Motorcycles	95	S10_1
9	10237	23	101.440	7	2333.120	2019-04-05	432	Shipped	Motorcycles	95	S10_1

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDERDATE	DAYS_SINCE_LASTORDER	STATUS	PRODUCTLINE	MSRP	PROI
2742	10350	20	112.220	15	2244.400	2019-12-02	2924	Shipped	Ships	54	
2743	10373	29	137.190	1	3978.510	2020-01-31	2865	Shipped	Ships	54	
2744	10386	43	125.990	4	5417.570	2020-03-01	2836	Resolved	Ships	54	
2745	10397	34	62.240	1	2116.160	2020-03-28	2810	Shipped	Ships	54	
2746	10414	47	65.520	9	3079.440	2020-05-06	2772	On Hold	Ships	54	

This is Head and Tail of the provided Sales Data

EDA cont..

```
(2747, 20)
```

The dataset has 2747 rows and 20 columns.

```
ORDERNUMBER            0
QUANTITYORDERED        0
PRICEEACH              0
ORDERLINENUMBER        0
SALES                  0
ORDERDATE              0
DAYS_SINCE_LASTORDER   0
STATUS                 0
PRODUCTLINE            0
MSRP                   0
PRODUCTCODE            0
CUSTOMERNAME           0
PHONE                  0
ADDRESSLINE1           0
CITY                   0
POSTALCODE             0
COUNTRY                0
CONTACTLASTNAME        0
CONTACTFIRSTNAME       0
DEALSIZE               0
dtype: int64
```

```
0
```

- No duplicates in the dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2747 entries, 0 to 2746
Data columns (total 20 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   ORDERNUMBER           2747 non-null   int64
 1   QUANTITYORDERED       2747 non-null   int64
 2   PRICEEACH             2747 non-null   float64
 3   ORDERLINENUMBER       2747 non-null   int64
 4   SALES                 2747 non-null   float64
 5   ORDERDATE             2747 non-null   datetime64[ns]
 6   DAYS_SINCE_LASTORDER  2747 non-null   int64
 7   STATUS                2747 non-null   object
 8   PRODUCTLINE           2747 non-null   object
 9   MSRP                  2747 non-null   int64
 10  PRODUCTCODE           2747 non-null   object
 11  CUSTOMERNAME          2747 non-null   object
 12  PHONE                 2747 non-null   object
 13  ADDRESSLINE1          2747 non-null   object
 14  CITY                  2747 non-null   object
 15  POSTALCODE            2747 non-null   object
 16  COUNTRY               2747 non-null   object
 17  CONTACTLASTNAME       2747 non-null   object
 18  CONTACTFIRSTNAME      2747 non-null   object
 19  DEALSIZE              2747 non-null   object
dtypes: datetime64[ns](1), float64(2), int64(5), object(12)
memory usage: 429.3+ KB
```

- All the Datatypes are correct no intervention is required.
- No Duplicates in the data
- No Missing values in the data

EDA cont..

- **Total Records**: 2,747 transactions

- **Order & Sales Metrics**:
  - *Average Quantity Ordered*: ~35 units per order
  - *Average Price Each*: ₹101.10
  - *Average Sales per Line Item*: ₹3,553
  - *Max Sales*: ₹14,082.80

- **Order Timing**:
  - *Order Dates*: Range from Jan 2018 to May 2020
  - *Days Since Last Order*: Mean ~1,757 days, with a wide spread (min 42, max 3,562)

- **Product & Customer Insights**:
  - *Top Product Line*: Classic Cars (949 entries)
  - *Most Frequent Product Code*: S18_3232
  - *Top Customer*: Euro Shopping Channel (259 orders)
  - *Top Order Status*: Shipped (2,541 orders)

- **Data Distribution**:
  - Most numeric fields like QUANTITYORDERED, PRICEEACH, and SALES show moderate variability, suggesting a diverse range of order sizes and values
  - MSRP and PRICEEACH are fairly aligned, indicating pricing consistency

	count	unique	top	freq	mean	min	25%	50%	75%	max	std
ORDERNUMBER	2747.000	NaN	NaN	NaN	10259.762	10100.000	10181.000	10264.000	10334.500	10425.000	91.878
QUANTITYORDERED	2747.000	NaN	NaN	NaN	35.103	6.000	27.000	35.000	43.000	97.000	9.762
PRICEEACH	2747.000	NaN	NaN	NaN	101.099	26.880	68.745	95.550	127.100	252.870	42.043
ORDERLINENUMBER	2747.000	NaN	NaN	NaN	6.491	1.000	3.000	6.000	9.000	18.000	4.231
SALES	2747.000	NaN	NaN	NaN	3553.048	482.130	2204.350	3184.800	4503.095	14082.800	1838.954
ORDERDATE	2747	NaN	NaN	NaN	2019-05-13 21:56:17.211503360	2018-01-06 00:00:00	2018-11-08 00:00:00	2019-06-24 00:00:00	2019-11-17 00:00:00	2020-05-31 00:00:00	NaN
DAYS_SINCE_LASTORDER	2747.000	NaN	NaN	NaN	1757.086	42.000	1077.000	1761.000	2436.500	3562.000	819.281
STATUS	2747	6	Shipped	2541	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PRODUCTLINE	2747	7	Classic Cars	949	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MSRP	2747.000	NaN	NaN	NaN	100.692	33.000	68.000	99.000	124.000	214.000	40.115
PRODUCTCODE	2747	109	S18_3232	51	NaN	NaN	NaN	NaN	NaN	NaN	NaN
CUSTOMERNAME	2747	89	Euro Shopping Channel	259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PHONE	2747	88	(91) 555 94 44	259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ADDRESSLINE1	2747	89	C/ Moralzarzal, 86	259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
CITY	2747	71	Madrid	304	NaN	NaN	NaN	NaN	NaN	NaN	NaN
POSTALCODE	2747	73	28034	259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
COUNTRY	2747	19	USA	928	NaN	NaN	NaN	NaN	NaN	NaN	NaN
CONTACTLASTNAME	2747	76	Freyre	259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
CONTACTFIRSTNAME	2747	72	Diego	259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
DEALSIZE	2747	3	Medium	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN

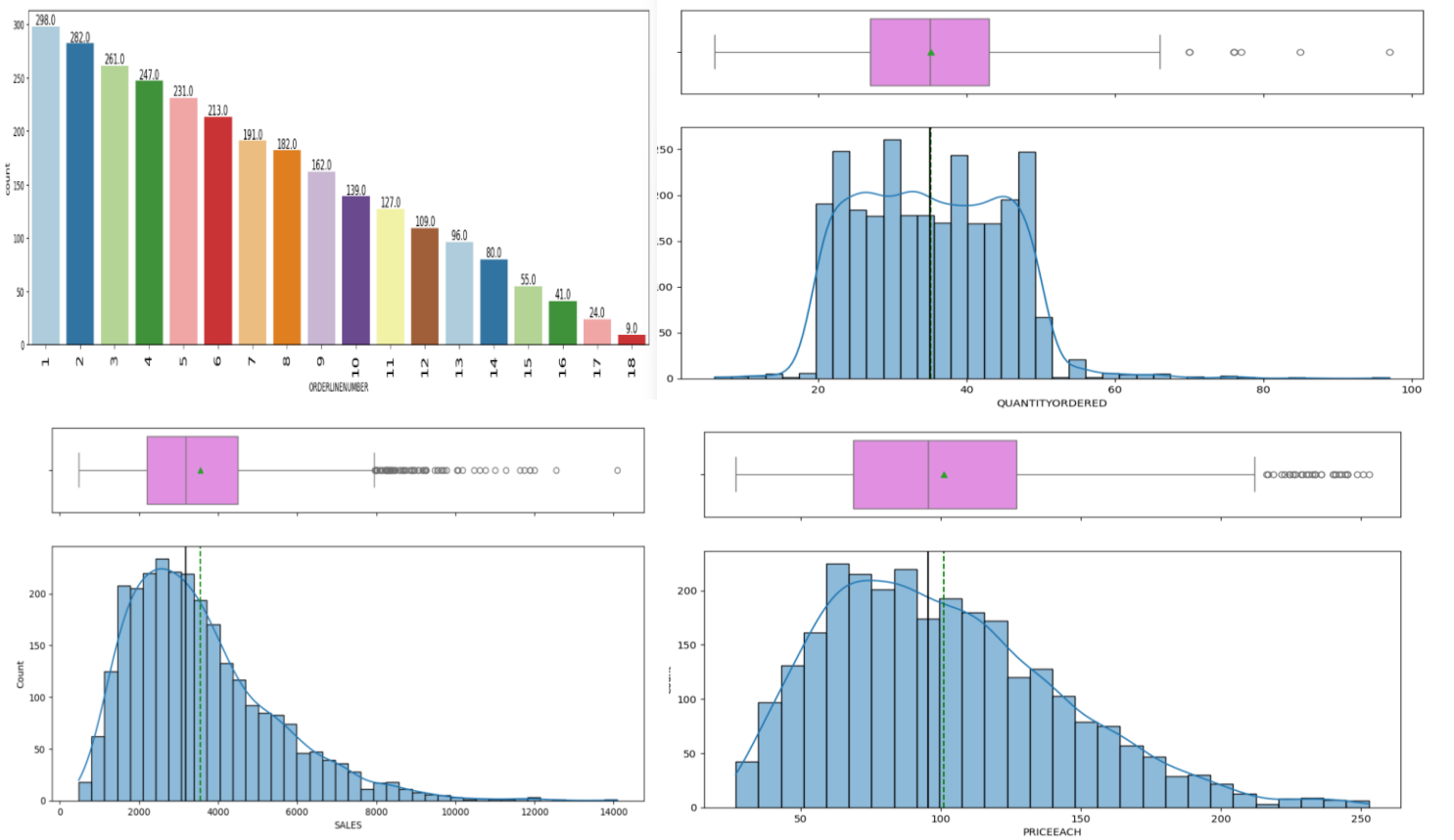
EDA cont..

•**OrderLineNumber:** Most orders include just a few items, with 70%+ of transactions involving top 1–5 product lines. Indicates focused purchasing behavior.

•**QuantityOrdered**: Median order quantity centers around 35 units. While most customers order within a consistent range, some place bulk orders.

•**Sales:** Sales per line-item average around ₹3,500. Most transactions are moderate in value, but high-value outliers.

•**PriceEach:** Product prices are tightly clustered near ₹100, with a few premium outliers. This stability suggests a standardized pricing model

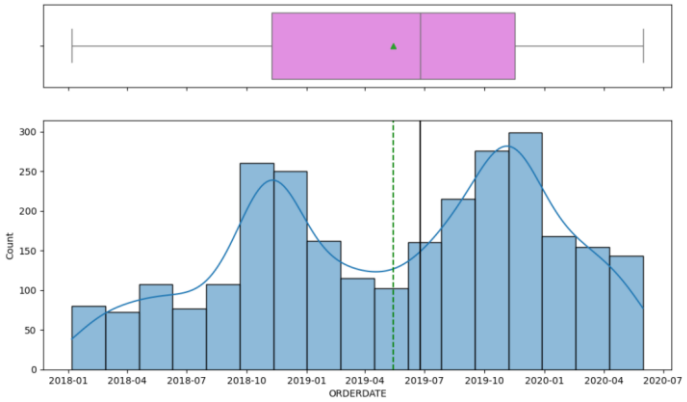


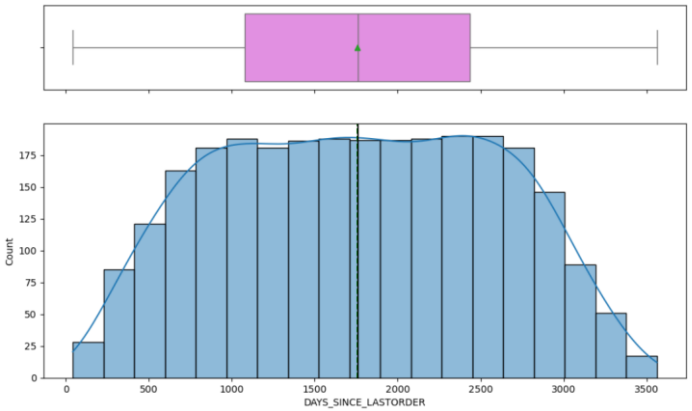
EDA cont..

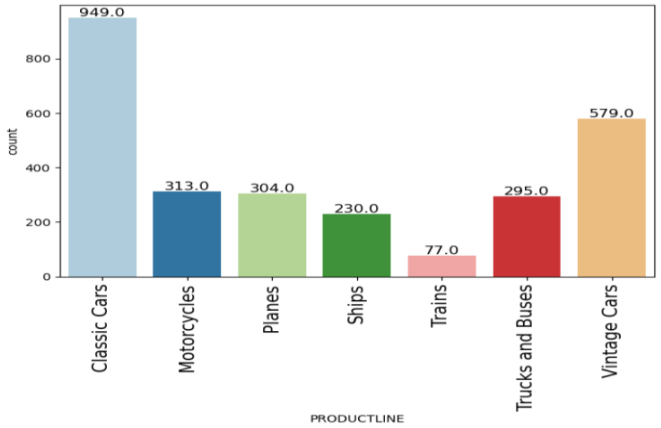
- **Order Trends Over Time:** Orders span from Jan 2018 to July 2020, with peak activity in April–July 2019, suggesting a period of increased demand.

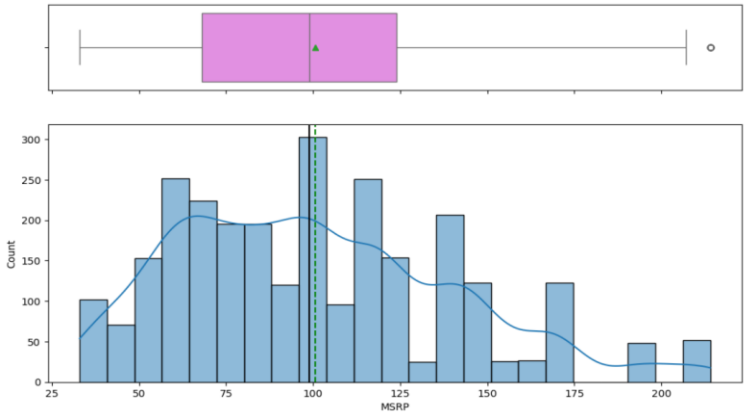
- **MSRP Distribution:** Most products have an MSRP clustered around ₹100, reflecting a standardized pricing strategy. A high outliers.

- **Product Line Performance:**
  - **Top Performer**: Classic Cars dominate with 949 orders.
  - **Strong Contenders**: Vintage Cars (579) and Motorcycles (313) maintain substantial demand, suggesting secondary growth areas.







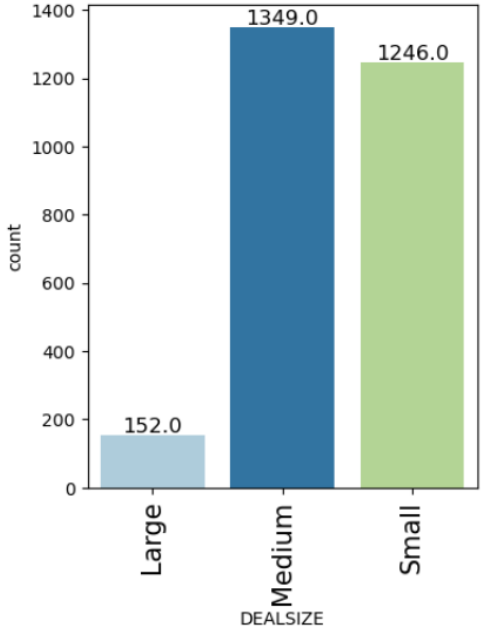


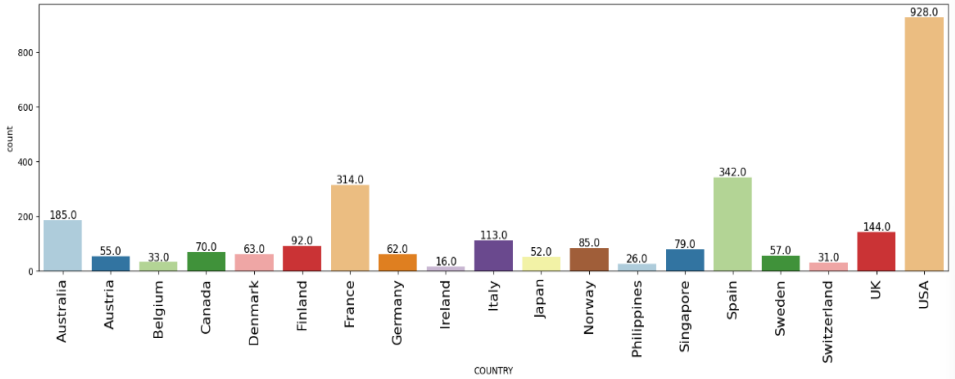
EDA cont..

- **Geographic Focus**: **USA dominates** customer count (928), followed by **Spain (342)** and **France (314)**—these are your core markets. Moderate reach in countries like **Australia, Finland, and Italy** suggests growth potential with region-specific strategies. Low-count regions such as **Ireland and Philippines** may need reevaluation or tailored penetration strategies.

- **Deal Size Dynamics: Medium (1,349)** and **Small (1,246)** deals form the business backbone—ideal for scalable, campaign-based marketing. **Large deals (152)**, though limited in volume, are likely **high-value**, justifying strategic relationship-building and loyalty programs.

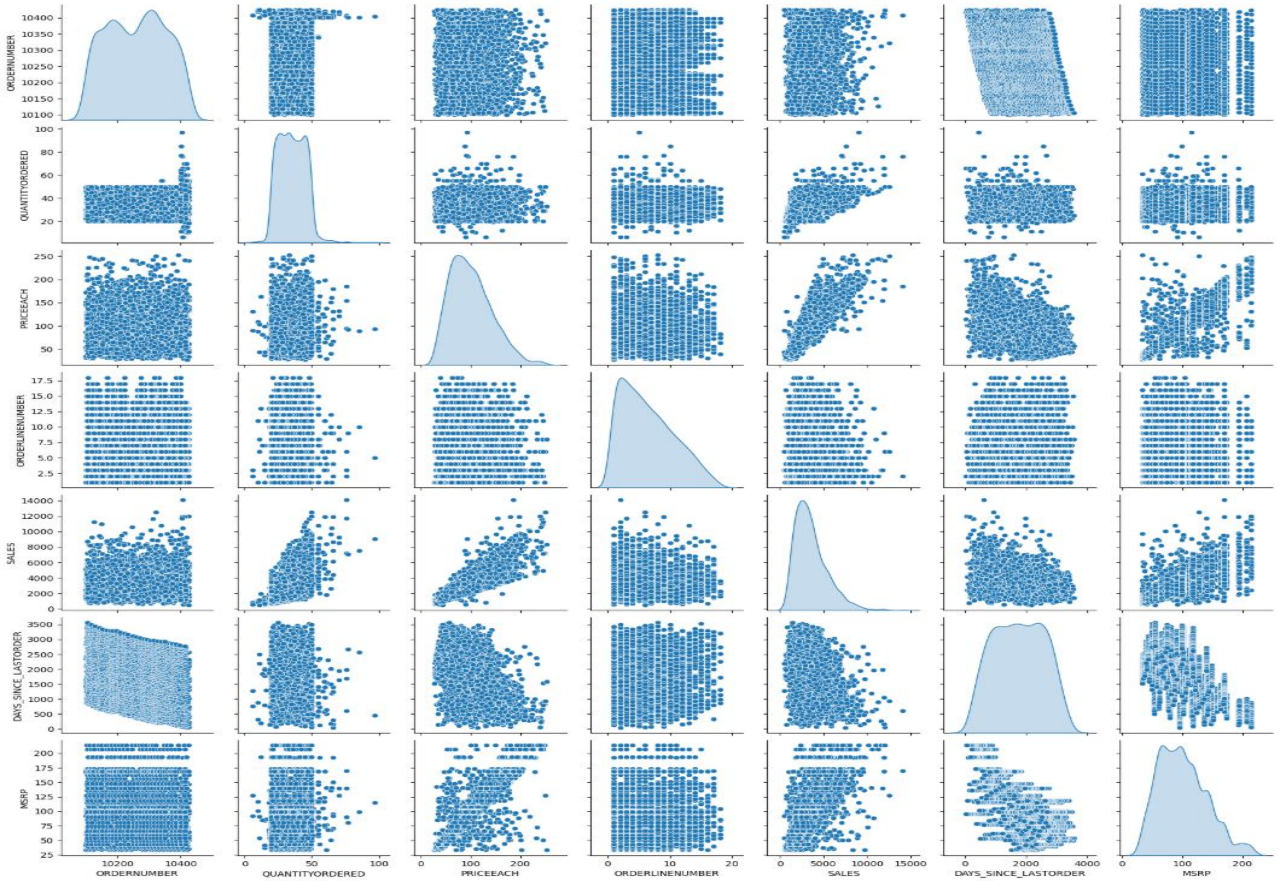
- **Key Correlations Driving Business Impact**:
  - **SALES strongly correlate with PRICEEACH (0.81)** and **MSRP (0.63)**—price optimization directly influences revenue.
  - **PRICEEACH ↔ MSRP correlation (0.78)** confirms consistent pricing strategy.
  - **DAYS_SINCE_LASTORDER is negatively correlated with SALES (-0.33)** and MSRP (-0.52), revealing that **longer gaps between purchases reduce revenue**—a compelling reason to implement re-engagement campaigns.





EDA cont..

- **SALES vs. PRICEEACH**: A strong positive correlation suggests that higher-priced items tend to generate more revenue.
- **SALES vs. MSRP:** A similar positive correlation implies that products with a higher MSRP are often sold at a higher price point.
- **QUANTITYORDERED vs. SALES:** If there's a noticeable trend where higher quantities lead to higher sales, bulk purchasing might be a key driver.
- **DAYS_SINCE_LASTORDER vs. SALES:** A weak correlation here might indicate irregular purchasing behaviors, signaling opportunities for retention-focused marketing strategies.
- Some histograms show right-skewed distributions (e.g., PRICEEACH and MSRP) indicating that most transactions involve lower-priced items, but occasional high-value orders skew the data.



EDA SUMMARY [INFERENCES]

1. Order & Purchase Behavior:

- Majority of transactions include just **1–5-line items**, suggesting focused customer intent and a strong case for bundling core products.
- **Quantity Ordered** centers around **35 units**, with occasional large orders—indicative of both routine buyers and high-volume clients.

2. Pricing & Revenue Patterns

- Both **Price Each** and **MSRP** cluster around **₹100**, reflecting pricing consistency, with a few premium items suitable for upselling.
- **Sales per line-item** averages ₹3,500, with outliers suggesting **high-value customers** that could be nurtured through loyalty programs.

3. Temporal Trends

- Orders span **Jan 2018 to July 2020**, peaking mid-2019—likely due to seasonal campaigns or demand surges.
- **DAYS_SINCE_LASTORDER** is broadly spread, revealing an inconsistent customer return cycle—pointing to the need for **re-engagement strategies**.

4. Product & Customer Segments

- *Classic Cars* dominate product orders, while *Trains* and *Ships* lag—valuable input for targeted promotions and inventory prioritization.
- **USA leads** customer count, followed by *Spain* and *France*—highlighting strong North American and Western European presence.

5. Deal Sizes & Value Segmentation

- Business is anchored by **Medium and Small deals** (97% combined), with **Large deals** offering strategic high-revenue potential.
- Geographic and product insights can guide market-specific targeting and volume-based incentives.

6. Correlations & Multivariate Patterns: Strong positive correlations:

- **Price Each ↔ Sales (0.81)**
- **Price Each ↔ MSRP (0.78)** → Pricing is a primary lever for revenue generation.
- **DAYS_SINCE_LASTORDER** shows **negative correlation with Sales**—the longer customers go inactive, the less they spend.

RFM Starts

Customer Segmentation using RFM analysis:

- **Number of Segments: 3**
  - **Justification:** Based on distinct RFM score groupings and customer behavior, the customers are segmented into **Active**, **At-Risk**, and **Inactive** categories. This trisection balances simplicity with strategic focus:
  - **Active**: High recency, frequency, and monetary value.
  - **At-Risk**: Moderate scores, with signs of potential churn.
  - **Inactive**: Low engagement across RFM dimensions.
  - This segmentation helps in tailoring marketing strategies with precision—targeting loyalty initiatives, re-engagement efforts, and reactivation campaigns respectively.

- **Parameters Used:** RFM model considers:
  - **Recency (R):** Days since the customer's last purchase (lower is better).
  - **Frequency (F):** Total number of transactions by the customer.
  - **Monetary (M):** Total sales revenue generated by the customer.

- **Assumptions Made:**
  - Higher scores on Frequency and Monetary represent more loyal, profitable customers.
  - Lower values on Recency indicate recent activity and thus, a higher likelihood of engagement.
  - Scoring was done on a scale (e.g., 1–3) for each RFM component, and summed or clustered to assign segment labels.

- **Recency:**
- *High (H)* → Customer purchased recently
- *Medium (M)* → Moderate time since last purchase
- *Low (L)* → Long time since last purchase
- **Frequency & Monetary:**
- *High (H)* → Frequent and/or high-value spender
- *Medium (M)* → Moderate purchase frequency/value
- *Low (L)* → Infrequent or low-value purchases

RFM (using KNIME)

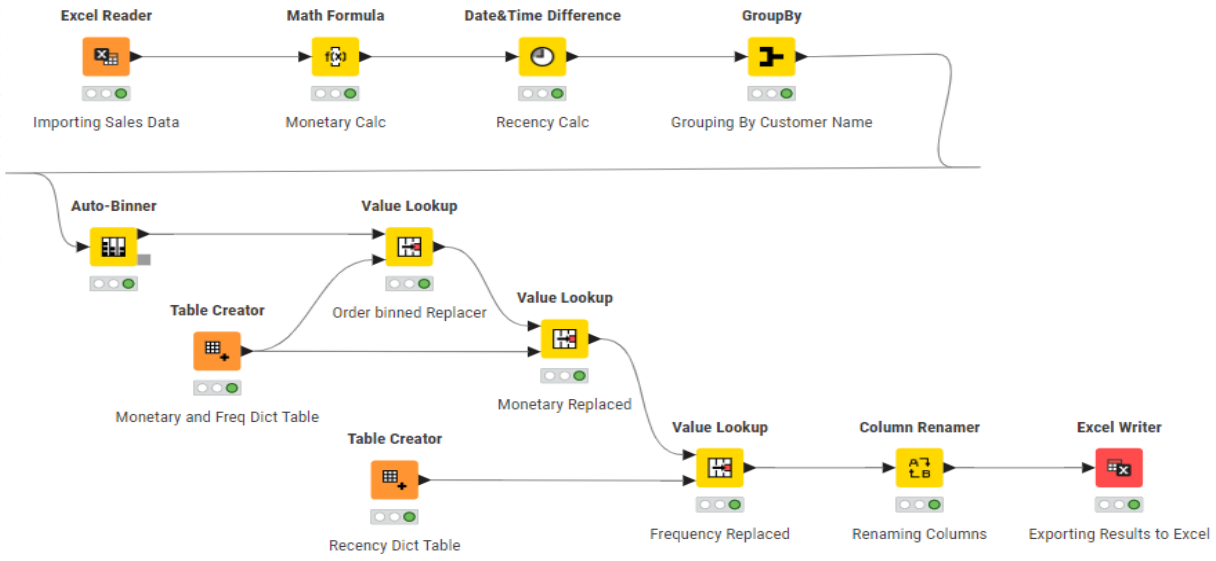
KNIME Workflow:

Output table

Rows: 89 | Columns: 28

#	RowID	CUSTOM... String	ORDERN... Number (inte...	QUANTIT... Number (dou...	PRICEA... Number (dou...	ORDERLI... Number (inte...	SALES Number (dou...	ORDERDA... Number (inte...	DAYS_SI... Number (dou...	STATUS String	PRODUC... Number (inte...	MSRP Number (dou...	PRODUC... String	PHONE Number (inte...	ADDRESS... Number (inte...	CITY String	POSTALC... Number (inte...	COUNTRY String	CONTAC... String
1	Row0	AV Stores, Co.	3	34.863	91.065	18	3,094.271	3	1,803.804	Shipped	4	92.843	S12_1108	51	51	Manchester	51	UK	Ashworth
2	Row1	Alpha Cognac	3	34.35	101.16	12	3,524.422	3	2,236.2	Shipped	4	97.15	S10_4757	20	20	Toulouse	20	France	Roulet
3	Row2	Amica Models &	2	32.423	110.853	17	3,619.895	2	1,318.615	Shipped	5	107.654	S10_1949	26	26	Torino	26	Italy	Accorti
4	Row3	Annà's Decorati	4	31.935	106.424	14	3,347.742	4	1,463.597	Shipped	4	104.717	S10_1949	46	46	North Sydney	46	Australia	O'Hara
5	Row4	Atelier graphiqu	3	38.571	92.239	4	3,454.28	3	1,424.429	Shipped	3	95.571	S10_2016	7	7	Nantes	7	France	Schmitt
6	Row5	Australian Colle	3	30.652	90.042	16	2,808.324	3	1,689.696	Shipped	4	88.13	S18_1342	23	23	Glen Waverly	23	Australia	Cornery
7	Row6	Australian Colle	5	35.018	104.59	15	3,654.462	5	1,698.782	Shipped	5	103.527	S10_1678	55	55	Melbourne	55	Australia	Ferguson
8	Row7	Australian Gift I	3	36.333	110.554	6	3,964.608	3	1,349.133	Shipped	6	111.533	S10_1949	15	15	South Brisbane	15	Australia	Callaghan
9	Row8	Auto Assoc. & C	2	35.389	99.488	17	3,601.907	2	1,266.278	Shipped	3	100.389	S10_1949	18	18	Versailles	18	France	Tonini
10	Row9	Auto Canal Peti	3	37.074	94.255	15	3,450.765	3	1,763.444	Shipped	2	94.852	S10_1678	27	27	Paris	27	France	Perrier
11	Row10	Auto-Moto Clas	3	35.875	92.8	4	3,309.908	3	2,281.75	Shipped	3	87.375	S18_3029	8	8	Brickhaven	8	USA	Taylor
12	Row11	Baane Mini Imp	4	33.812	108.574	16	3,643.725	4	1,805.75	Shipped	5	107.469	S10_1678	32	32	Stavern	32	Norway	Bergulfsen
13	Row12	Bavarian Collec	1	28.643	84.289	14	2,499.566	1	2,239.357	Shipped	3	82.714	S18_1662	14	14	Munich	14	Germany	Donnermeyer

DEALSIZE String	Monetary Number (dou...	Recency Number (long)	ORDERN... String	Monetary... String	Recency [... String	Freqency... String	Monetary... String	Recency_... String
Medium	157,807.81	196	Bin 2	Bin 3	Bin 2	M	H	M
Medium	70,488.44	64	Bin 2	Bin 1	Bin 1	M	L	H
Large	94,117.26	265	Bin 1	Bin 2	Bin 3	L	M	L
Small	153,996.13	83	Bin 3	Bin 3	Bin 2	H	H	M
Medium	24,179.96	188	Bin 2	Bin 1	Bin 2	M	L	M
Medium	64,591.46	22	Bin 2	Bin 1	Bin 1	M	L	H
Medium	200,995.41	184	Bin 3	Bin 3	Bin 2	H	H	M
Large	59,469.12	119	Bin 2	Bin 1	Bin 2	M	L	M
Large	64,834.32	233	Bin 1	Bin 1	Bin 3	L	L	L
Medium	93,170.66	54	Bin 2	Bin 2	Bin 1	M	M	H
Medium	26,479.26	180	Bin 2	Bin 1	Bin 2	M	L	M
Medium	116,599.19	208	Bin 3	Bin 2	Bin 2	H	M	M
Medium	34,993.92	259	Bin 1	Bin 1	Bin 3	L	L	L



RFM (using KNIME) Inferences

Count of CUSTOMERNAME		Monetary_HLM				
Recency_HLM	Freqency_HLM	H	M	L	Grand Total	
H	H	9	2		11	Active Customers
	M	1	9	2	12	
M	H	6	3	1	10	At-Risk Customers
	M	5	16	5	26	
	L		5	3	8	
L	M	1	1		2	Inactive Customers
	L		8	12	20	
Grand Total		22	44	23	89	

Insights

- **Active Customers (23 Total):**
  - Gold Tier (9 customers): These are your most loyal and profitable segment. They've purchased recently, frequently, and in high amounts.
  - Silver Tier (12 customers): Still valuable—slightly lower on one or two RFM dimensions but show strong potential.
  - Bronze Tier (2 customers): Actively buying, but either less frequently or with smaller purchase values.
- **At-Risk Customers (44 Total)**
  - Large concentration in Silver (24) and Bronze (14): These customers were previously engaged but haven't purchased recently.
  - Gold (6 customers): Once very valuable—if they're starting to lapse, this is a red flag worth immediate attention.
- **Inactive Customers (22 Total)**
  - Mostly Bronze (20 customers): These likely had infrequent, low-value transactions and have since disengaged.
  - Silver (2 customers), Gold (0): Indicates minimal high-potential customers have become completely inactive.

HHH
HMH & HMM
HHL & HML & HLH & HLM & HLL

Active Customers	
Gold	9
Silver	12
Bronze	2

MHH
MMH & MMM
MHL & MML & MLH & MLM & MLL

At-Risk Customers	
Gold	6
Silver	24
Bronze	14

LHH
LMH & LMM
LHL & LML & LLH & LLM & LLL

Inactive Customers	
Gold	0
Silver	2
Bronze	20

RFM (using KNIME) Inferences

Recommendations:

- Nurture Active Customers
  - Gold Tier: Offer exclusive loyalty perks, sneak peeks at new products, or early-bird deals.
  - Silver Tier: Encourage upselling and cross-selling with personalized product bundles.
  - Bronze Tier: Use satisfaction surveys or onboarding-style communication to build stronger habits and move them up.
- Win Back At-Risk Customers
  - Send "We miss you" emails with time-sensitive promotions.
  - Re-target them via ads showing products similar to past purchases.
  - Offer tier-based incentives to encourage re-engagement—perhaps a one-time boost to reach Silver or Gold status.
- Revive Inactive Customers
  - Launch a reactivation campaign with bold offers (e.g., 25% off next order) or fresh messaging focused on new arrivals.
  - Use feedback forms to identify pain points that may have caused disengagement.
  - Focus retention resources on Bronze-level customers only if acquisition costs are lower than potential return.